Mock Interview: BigTable

Test your knowledge of BigTable's architecture by answering these questions.

Quiz ?

Statement

How does BigTable scale?

Answer

BigTable breaks a table into a number of Tablets, where each Tablet contains data associated with a contiguous range of rows. Initially, each table consists of just one table. As the table grows, it is split into multiple Tablets. BigTable master is responsible for assigning Tablets to Tablet servers. Tablet servers are responsible for handling read and write requests to the Tablets that they are assigned as well as splitting Tablets that have grown too large. It is very easy to add new Tablet servers in a BigTable cell. Any newly added Tablet server is assigned Tablets either from the pool of unassigned Tablets or from other Tablet servers.

Hide Answer

Quiz ?

Statement

How does BigTable ensure fault-tolerance and reliability of data?

Answer

BigTable uses GFS, which replicates data to multiple ChunkServers for reliability and fault tolerance. BigTable master is responsible for detecting Tablet server failures. It does this by periodically checking the status of the Chubby lock against each Tablet server. When the master finds out that a Tablet server has gone dead, it reassigns the tablets of the failing Tablet server.

Hide Answer

Quiz ?

Statement

How does BigTable ensure durability?

Answer

BigTable ensures durability by replicating data across multiple replicas through GFS, which stores the data on the disk.

Hide Answer

Quiz ?

Statement

What BigTable components can become Single Point of Failure (SPOF), and how does BigTable handle them?

Answer

Here are the BigTable components that are SPOF. BigTable master BigTable uses Chubby to ensure that there is only one active master. If Chubby becomes unavailable for an extended period of time, BigTable will also become unavailable. When the master dies, Google's Cluster Management System starts a new master. BigTable master is not involved in the core task of mapping tablets onto the underlying files in GFS (this is handled by Tablet servers). This means that BigTable clients do not have to communicate with the master at all. This significantly reduces the load on the master and enables clients to read data even when the master is not available. Chubby master Chubby becomes unavailable if its master dies. Chubby uses PAXOS to elect a new master from other replicas. The new master loads the system state from the database and the transaction log before serving requests. GFS master GFS cannot work without its master. GFS relies on an external monitoring infrastructure to detect the master failure and switch the traffic to the backup master server. GFS uses Operation Log and Checkpointing to save the current master state. A newly elected master may load the checkpoint into memory, replay any subsequent operations from the Operation Log, and start serving the traffic.

Hide Answer

Quiz ?

Statement

Does BigTable become unavailable if its master is down?

Answer

The master is responsible for all metadata operations and other control operations like monitoring and load balancing Tablet servers. Therefore, if the master is down, the cluster will become unhealthy, and clients will not be able to perform any schema changes. During this time, clients can still access data as the data transfer happens directly between clients and Tablet servers.

Hide Answer

Quiz ?

Statement

How does BigTable perform load-balancing of Tablet servers?

Answer

BigTable master is responsible for assigning Tablets to Tablet servers and thus maintains a global view of the system. Also, the master periodically asks Tablet servers about their current load and based on this information, moves Tablets around to balance the load between Tablet servers.

Hide Answer

Quiz ?

Statement

How does BigTable manage the separation between the data and control flow?

Answer

BigTable maintains a strict separation between control and data flow. BigTable clients talk to the Master for all metadata operations, whereas all data access happens directly between the Clients and the Tablet servers.

Hide Answer

← Previous
Quiz: BigTable

Next →
Introduction: System Design Patterns

☐ Mark as Completed